Extracting Glacier Calving Fronts by Deep Learning: The Benefit of Multispectral, Topographic, and Textural Input Features

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***Abstract*— An accurate parameterization of glacier calving is essential for understanding glacier dynamics and constraining ice-sheet models. The increasing availability and quality of remote sensing imagery open the prospect of a continuous and precise mapping of relevant parameters, such as calving front locations. However, it also calls for automated and scalable analysis strategies. Deep neural networks provide powerful tools for processing large quantities of remote sensing data. In this contribution, we assess the benefit of diverse input data for calving front extraction. In particular, we focus on Landsat-8 imagery supplementing single-band inputs with multispectral data, topography, and textural information. We assess the benefit of these three datasets using a *dropped-variable* approach. The associated reference dataset comprises 728 manually delineated calving front positions of 23 Greenland and two Antarctic outlet glaciers from 2013 to 2021. Resulting feature importance emphasizes both the potential integrating additional input information as well as the significance of their thoughtful selection. We advocate utilizing multispectral features as their integration leads generally to more accurate predictions compared with conventional single-band inputs. This is especially prevalent for challenging ice mélange and illumination conditions. In contrast, the application of both textural and topographic inputs cannot be recommended without reservation, since they may lead to model overfitting. The results of this assessment are not only relevant for advancing automated calving front extraction but also for a wider range of glaciology-related land surface classification tasks using deep neural networks.**

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I. Introduction

CALVING of marine terminating outlet glaciers results in seasonal and interannual changes of their frontal positions. These calving front variations are both triggered by changes in the glacier dynamics, but also affect the dynamics of tidewater glaciers [1], [2], [3]. The accurate detection of calving rates is crucial for investigating the physical mechanism and controlling factors [4], [5], [6], [7], [8]. Moreover, accurately representing calving front dynamics in models is an essential part of constraining glacial evolution [9] and can improve the simulations of mass loss and projections of future sea-level contributions [10], [11], [12], [13], [14]. Consequently, temporally, and spatially comprehensive calving front datasets are essential for a better understanding and modeling of marine terminating glaciers.

The increasing availability and quality of both optical as well as radar satellite imagery enable us to realize a continuous mapping of calving front locations. However, most current calving front datasets [15], [16], [17], [18], [19], [20], [21] are based on manual delineation, which is laborious and infeasible considering the immense data volume. Therefore, calving front records are often limited in temporal resolution making seasonal analyses difficult. The huge amount of data accentuates the necessity for automated and scalable delineation strategies. Driven by this issue, various empirical feature extraction algorithms have been developed in the last decades pursuing robust automated calving front extraction [22], [23], [24], [25], [26], [27]. However, most of these methods require case specific modifications and are not tested for spatial transferability and large-scale applications.

As an alternative to these established image processing techniques and owing to advances in the field of machine learning, deep artificial neural networks (ANNs) are becoming the model of choice for solving complex image processing tasks. Mohajerani *et al.* [28] introduced this approach by mapping calving front locations of three Greenlandic outlet glaciers using single-band optical Landsat imagery emphasizing its performance, especially against hardcoded edge detection operators. Cheng *et al.* [29] substantially expanded

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this framework by advancing preprocessing procedures, the underlying neural network architecture, as well as achieving a robust handling of Landsat-7 scanline correction errors. Other studies focused on the usage of synthetic-aperture radar (SAR) imagery, where both single-polarization inputs [30] and dual-polarization inputs [31] have been used. While the usage of SAR has clear advantages, such as the independence from weather and illumination, it also imposes some challenges. Particularly, the backscatter characteristics of glacial ice vary throughout the year, making it hard to distinguish between relevant surfaces. Heidler *et al.* [32] addressed this problem by intertwining semantic segmentation with edge detection into a single model. They achieved better prediction performance under these problematic conditions. Considering both optical and radar sensor characteristics, Zhang *et al.* [33] developed a more generalized framework capable of processing multisensor imagery.

Despite their similarities, all these approaches differ in study areas, input data, preprocessing strategies, and neural network architecture. Recently, Heidler *et al.* [32] and Zhang *et al.* [33] provided extensive comparisons between different deep learning models and architecture modifications. However, only little consensus and literature exist regarding input feature selection and their specific contribution to prediction performance. Since remote sensing data are usually multimodal and ANN excel at synergistically integrating diverse datasets [34], we expect high potential on including additional input information. Hence, in this contribution, we make an important step proposing to include further input features for automated calving front delineation from optical Landsat-8 imagery. We focus on the following three input datasets: multispectral bands, topographic model data, and textural features. The motivation for utilizing these three datasets is given by their significance for a wide range of remote sensing tasks.

Section II covers utilized datasets and the scope of our work. Section III explains the setup and applied methods. Section IV contains the treatment of results and a subsequent discussion. Finally, Section V gives a summary and outlook.

II. Data and Scope

A. Data Source

The analysis made in this article is based on optical Landsat-8 imagery. In particular, we utilize the radiometrically calibrated and orthorectified level-1 data products provided by the United States Geological Survey (USGS). With its two imaging sensors, operational land imager (OLI) and thermal infrared (TIR) sensor, Landsat-8 provides wide multispectral capabilities ranging from visible (VIS) over near-infrared (NIR) and short wave infrared (SWIR) to TIR wavelengths. In total, Landsat-8 has 11 different bands. Besides the 30-m resolution of the reflective bands and 100-m resolution of the thermal bands, an additional panchromatic band combines visible wavelengths into one channel achieving 15-m resolution. With the exception of Band 9, which lies outside an atmospheric window and is, therefore, designed for observing the atmosphere, all available bands are used. In itself, Landsat-8 has reasonable spatial and temporal coverage, achieving subweekly revisit times of all Greenlandic glaciers outside polar night. A combination with the recently launched Landsat-9 decreases revisit intervals even further, reaching subdaily sampling for north Greenland glaciers outside polar night [35]. In the following, we introduce three particular input datasets we apply in this study.

Most optical satellite-sensors capture image data within specific wavelength ranges across the electromagnetic spectrum. Utilizing these multispectral bands or band combinations is widely established for glacier mapping and monitoring tasks [36]. The concept is based on the distinct reflective curves of different surface types along the electromagnetic spectrum. Fig. 1 illustrates these reflectance curves for different surfaces on and around Greenland glaciers along with the atmospheric transmission and Landsat-8 band designations used in this study. Fig. 2 displays these surface dependent reflectivities exemplarily for three Greenlandic glaciers: [Fig. 2(a)–(d)] Store Glacier, [Fig. 2(e)–(h)] Daugaard Jensen Glacier, and [Fig. 2(i)–(l)] Jakobshavn Isbræ. The VIS spectrum (Bands 1–4 and 8) is characterized by high reflectance of glacier ice and ice mélange as well as very high reflectance of snow [37]. Band 1 senses deep blues and violets, enabling a better view of shady surfaces than longer wavelengths. The NIR reflectivity is decreased by both grain size and the presence of liquid water [38]. The latter facilitates a better spectral separability between glacier and ice mélange. SWIR wavelengths are marked by a high reflectance of bedrock and a low reflectance of snow and ice. Signals in the TIR spectrum are influenced by thermal processes related to both surface and weather conditions [39], [40]. Depending on these conditions and the responding surface temperatures, TIR reflectivities of snow, ice, and water may be very similar or noticeably different. Although some studies successfully applied multispectral data to facilitate automated glacier mapping [41], [42], [43], current approaches for ANN-based calving front extraction are limited to the use of single-band inputs, discarding additional multispectral information.

As a second dataset, we assess topographic elevation model data. Baumhoer *et al.* [31] and Heidler *et al.* [32] already experimentally utilized a digital elevation model (DEM) and found that a careful integration facilitates more accurate model predictions. This is especially the case under challenging conditions, where synchronous topography information helps to constrain the glacier front position relative to the fjord topography. Nevertheless, Heidler *et al.* [32] also observed a severe model overfitting on the DEM for dynamic regions where elevation data and satellite imagery are contradictory. This resulted in systematic errors at the glacier front and ultimately to an accuracy decrease compared with models without DEM inputs. To counteract this, we only make use of above sea-level bed elevation. Bed elevation data have the ice thickness subtracted and are, therefore, temporally stationary. Thus, we avoid overfitting at the glacier front while still providing topographical context for the model. By disregarding the sub-sea-level topography, specifically by setting negative elevation values to zero, we create an ice-free DEM-like

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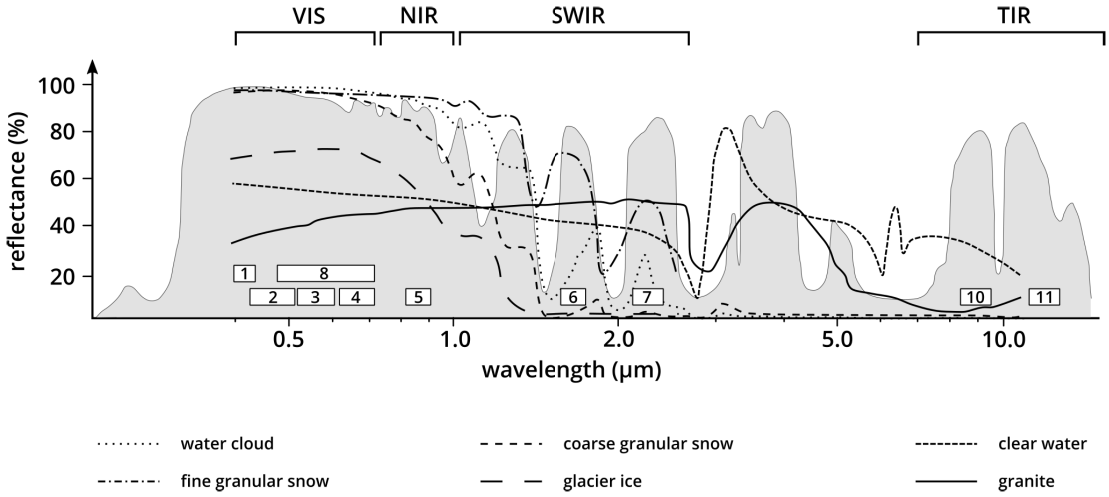


Fig. 1. Atmospheric transmission, Landsat-8 band designations (numbers), and typical reflectance curves for different surfaces found on or around glaciers. Figure adapted from Kääb *et al.* [36].

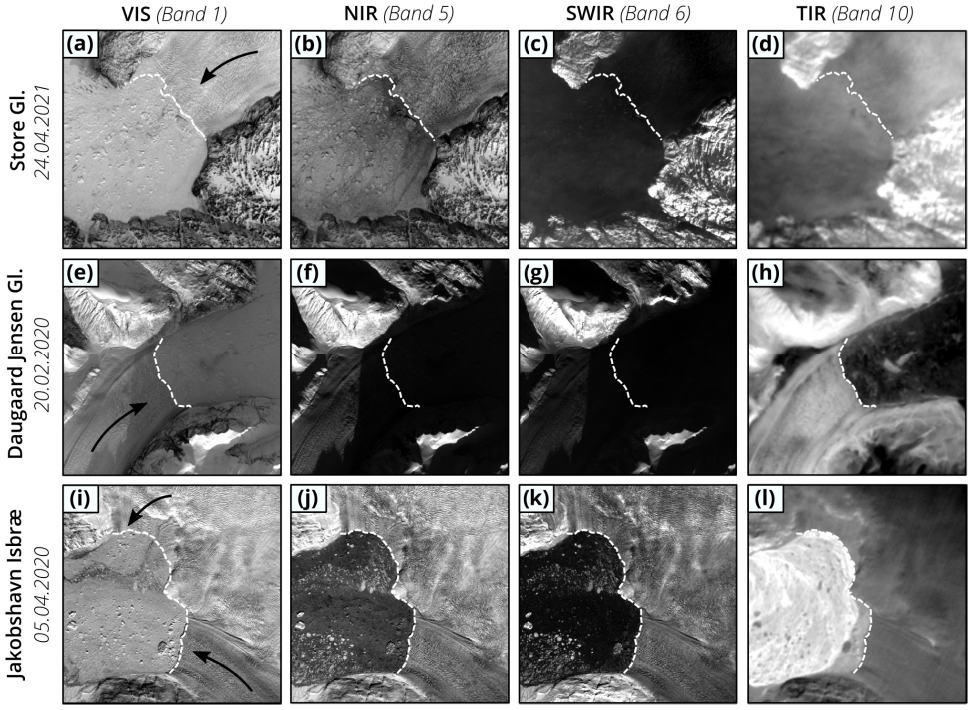


Fig. 2. Surface dependent reflectivities for (a)–(d) Store Glacier, (e)–(h) Daugaard Jensen Glacier, and (i)–(l) Jakobshavn Isbræ for different wavelength ranges of Landsat-8, namely, the deep blue (Band 1), NIR (Band 5), SWIR (Band 6), and TIR (Band 10). Arrows indicate ice flow direction. The calving front is highlighted as a dashed line. For the location of the specific glaciers, see Fig. 4.

product. Bed topography data are taken from BedMachine Greenland v3 [12].

Satellite imagery of polar regions, especially in Greenland, consists mostly of natural textures. Hence, our satellite images are stochastic in nature and are suitable for methods of statistical image analysis. For our third dataset, we analyze the gray-level co-occurrence matrix (GLCM), which counts the co-occurrence of neighboring gray levels in the image, enabling a statistical extraction of textural features [44]. Applying a 9- × 9-px sliding window on the 15-m resolution panchromatic band, we extract the following textural features: angular second moment, contrast, entropy, dissimilarity, homogeneity, and correlation. The idea in utilizing these features is that the homogeneous texture of the ice mélange, water, or fast ice stands out to the glacier front, which is characterized by crevasses, seracs, and other distinguishing features [25]. Fig. 3 emphasizes this aspect by showing [Fig. 3(a)] the panchromatic image, [Fig. 3(b)–(g)] spatial characteristics of GLCM textural features, and [Fig. 3(h)] the topography model exemplarily for Hagen Bræ in North Greenland (Fig. 4). Although the textural features allow for a better separability between different surface types, they do

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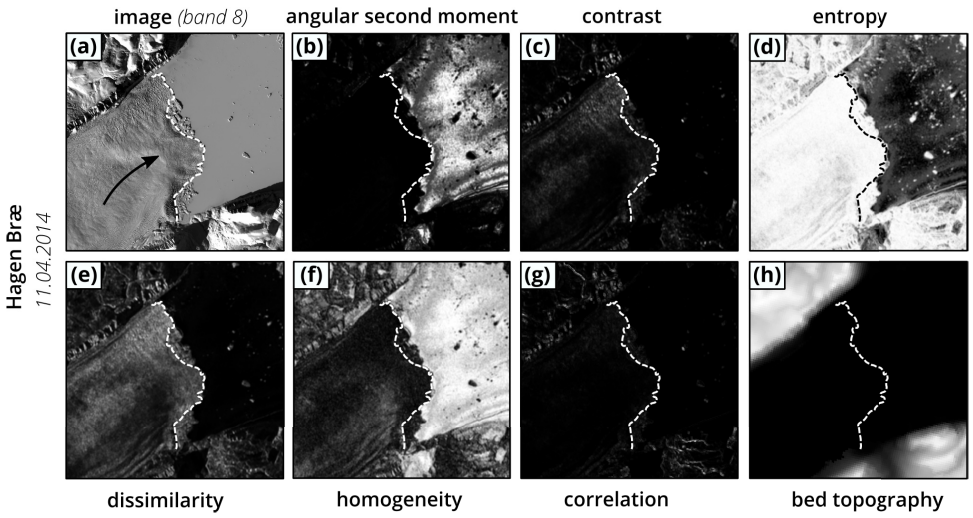


Fig. 3. (a) Panchromatic image, (b)–(g) textural features, and (h) above sea-level topography of Hagen Bræ. Arrows indicate ice flow direction. The calving front is highlighted as a dashed line.

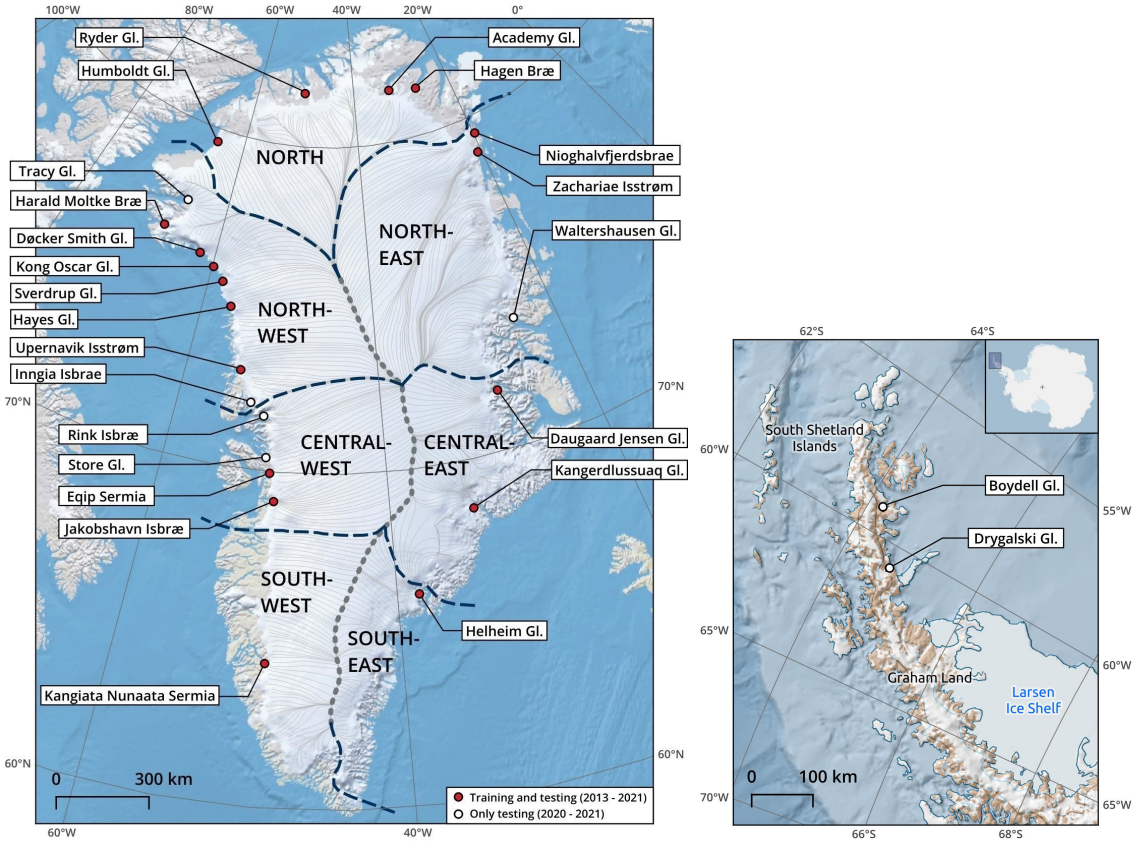


Fig. 4. Geographic location of the 23 Greenlandic and the two Antarctic outlet glaciers used in our reference dataset. Red dots indicate glaciers used for training and testing. Glaciers marked with a white dot are only used for model testing. (a) Greenland. (b) Antarctic Peninsula.

not well represent objects or areas smaller than the selected window size. In Fig. 3, this is particularly evident for the small icebergs near the glacier front. Reducing the window size further would enable a sharper transition between relevant surface types, but also leads to a noisy and unstable feature extraction. Since the GLCM feature transformation is highly nonlinear, it cannot be learned easily by neural networks and can, thus, provide a valuable enrichment of the input data. Combining GLCM textural features with ANNs has, therefore, already been explored in various areas of image

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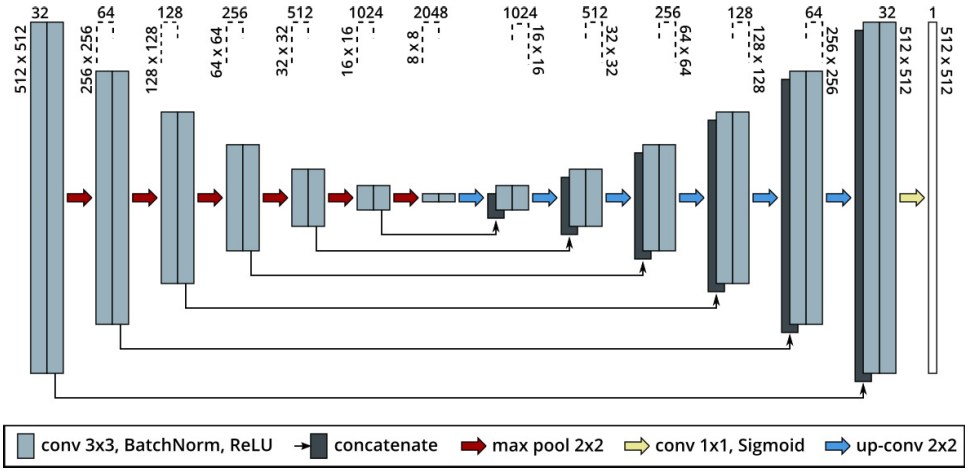


Fig. 5. U-Net-based processing architecture used in this study. A contracting sequence consists of convolutions followed by batch normalization, a rectified linear unit (ReLU), and a max-pooling operation. Expanding sequences are composed of a concatenation with spatial context from the contracting path, convolutions followed by batch normalization and an ReLU. Note that all dimensions are denoted at the top of the blocks.

analysis [45], [46], [47]. Here, we will assess how this method can be applied for the calving front extraction.

B. Reference Dataset

Over the last two decades, the Greenland ice sheet has experienced an increased ice-mass loss, which is caused mainly due to increased surface melt and ice discharge [48]. Along with accelerating glacier velocity and frontal ablation, high retreat rates have been found [2]. The quantification and understanding of glacier dynamics will provide important insights into the Greenland ice sheet response to ongoing change as well as long-term climate forcing. To facilitate these analyses and resolve spatial and temporal data gaps of calving front positions, we choose the Greenland ice sheet as the main study area for our assessment.

Our reference dataset consists of 18 Greenland outlet glaciers, which are used for model training, validating, and testing. Additional five Greenlandic glaciers and two glaciers at the Antarctic Peninsula are used for model testing, specifically for assessing spatial transferability. Fig. 4 gives an overview of the spatial distribution of our analyzed glaciers. The selection ensures the diversity of morphological features as well as of calving and ocean conditions. As these conditions also change throughout the year with cloud cover and illumination changing scene by scene, we have to ensure sufficient sampling during the observation periods from spring to autumn. In total, the reference dataset consists of 728 manually delineated calving front positions, of which 585 calving fronts (from 2013 to 2019) are used as input for model training, and 143 calving fronts (from 2020 to 2021) are used for model testing.

III. Methods

A. Neural Network Implementation

Mapping glacier calving front locations is a contour detection task. So far, the most popular approach in accomplishing this with ANNs is based on pixel-wise semantic image segmentation using a convolutional neural network followed by vectorizing the output mask. In particular, U-Net-type architectures [49] have proven to be very effective for various study areas and sensor types [28], [30], [31], [32]. Since this study focuses on input data assessment rather than advancing ANN architecture, we adapt the already established U-Net model for our analysis. The U-Net itself is based on a contracting path reducing spatial information while increasing feature information followed by an expanding path where spatial and feature information is combined by alternating concatenations and deconvolution. The receptive field of a U-Net is defined by the number of contracting and expanding blocks. Deeper models have larger receptive fields and, thereby, more spatial context information for each pixel classification. Since calving front delineation benefits significantly from increased spatial context [32], our model is expanded by two additional resolution levels, i.e., from four to six levels. Fig. 5 shows the processing architecture and relevant dimensions of the U-Net used in this study.

The presented model performs a land surface classification where a land and glacier class is semantically segmented from a water class, with the calving front later to be extracted from the boundary between these two classes. The ANN input consists of the multispectral, topographic, and textural datasets described in Section II. Raster subsets have the dimensions of 512×512 px and are centered at the calving front location of the corresponding glacier. With a uniformly applied 30-m ground sampling distance, each input raster subset covers an area of about 15×15 km. To counteract overexposed areas in our satellite imagery, we apply additional image enhancement in the form of a cumulative count cut clipping the data between the 0.1% and 98% of each multispectral band. In total, we use 17 input layers, namely, ten Landsat-8 bands, six GLCM textural features, and bed topography. Finally, all of these 17 input layers are normalized to the range between 0 and 1 using an 8-bit quantization.

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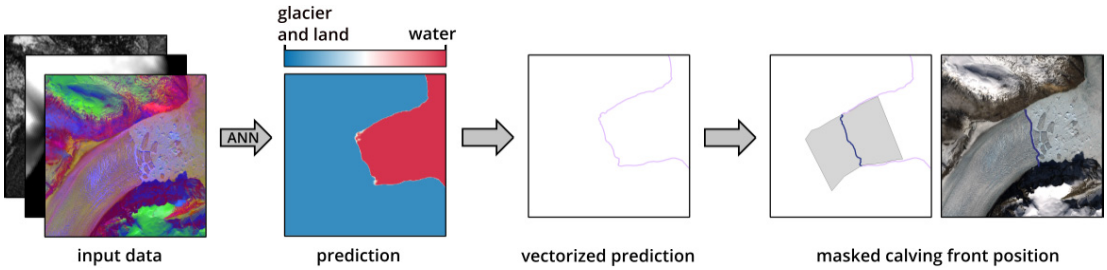


Fig. 6. Overview of the processing workflow using an exemplary satellite scene of the Daugaard Jensen Glacier. Through an ANN, a land and glacier class is semantically segmented from a water class. After vectorization, we extract the calving front using a static mask.

To counteract model overfitting, we select every fifth image of the training data, a total of 164, for internal validation. The remaining training data are augmented eightfold by flipping and rotating. The resulting 5232 raster subsets are finally used for training the model with randomized batches of size 8. For this, we apply the Adam optimization algorithm [50] on a binary cross-entropy loss function for 150 epochs. The final model weights are selected based on the binary classification accuracy of the internal validation dataset. The ANN implementation is realized using the TensorFlow 2.3 library [51]. Training is carried out on an IBM Power 9 node using a NVIDIA V100 GPU with 32-GB high bandwidth memory. With each epoch taking 3–4 min, our model is trained after 7.5–10 h. Up to 32 models can be trained in parallel.

Fig. 6 gives a basic overview of the processing workflow. The trained model performs an automated land surface classification, where each image pixel is denoted a probability between 0 (water) and 1 (glacier and land). As part of postprocessing, this floating point number probability mask is vectorized using the Geospatial Data Abstraction Library contour algorithm [52], thresholding at a probability of 0.5. Larger glaciers, with calving fronts exceeding the 512- $\times$ 512-px input window, are separated into several independent predictions, which are averaged in the overlapping area before vectorization. This concept is adopted from Baumhoer *et al.* [31]. It is applied for Humboldt Glacier, Nioghalvfjerdsbræ, and Zachariae Isstrøm, which are split into seven, three, and two separate overlapping predictions, respectively. Finally, we extract the glacier calving front by intersecting the vectorized coastline with a static mask. This mask is manually created for each glacier and specifies the corridor of possible calving front locations.

B. Accuracy Assessment

To validate the ANN model, it was applied to the test dataset introduced in Section II-B. This test dataset contains 177 manually delineated frontal positions for all Greenlandic glaciers used for training the model, but for the different time periods 2020 and 2021. Also, included were an additional five Greenlandic glaciers and two glaciers at the Antarctic Peninsula.

As a main error metric, we choose the distance between the predicted and the manually delineated calving front trajectory. Conceptually, this corresponds to the area between the two curves normalized by their length. We implement this metric by averaging the minimal distance to the manual delineation every 30 m along the predicted front. We define two different accuracy estimates based on the nonnormal distribution of the averaged distance error for the 177 test scenes.

1) The *mean distance error* is sensitive toward outliers, which are mostly due to large misclassified areas and, therefore, falsely extracted calving front positions of scenes with very challenging conditions.
2) The *median distance error*, in contrast, is not sensitive toward individual scenes. It gives information on the general scene-by-scene performance as well as on a systematic overfitting of our model.

Although our model is fit using pixel-wise binary cross entropy, the mean and median distance error metrics are more representative measures of model performance. This is due to the ANN-based pixel-wise classification only being an intermediate step in our processing scheme. A high binary classification accuracy does not necessarily translate into an accurate calving front prediction, as binary classification accuracy is primarily important directly at the glacier front.

C. Assessment of Feature Importance

Deep learning has proven to be a powerful set of tools in many data-driven fields. However, these methods often produce complex models that provide little to no insight how these predictions are derived. Numerous methods are attempting to look inside this "black box" function and to understand the contribution of a feature to the predictive power of a model. Popular and computational convenient methods are based on the *permute-and-predict* strategy first introduced by Breiman [53]. Since our model inputs exhibit significant statistical dependences, which is a common property of multilayer remote sensing imagery, diagnostics based on this *permute-and-predict* approach can be highly misleading [54], [55], [56]. Hooker and Mentch [57] explain this behavior in detail and suggest several alternatives generally based on either conditional permutation or model relearning. Taking the results of these studies into account, we evaluate our features using the *dropped variable* concept suggested by Hooker and Mentch [57], which is equivalent to the *leave one covariate out* framework introduced by Lei *et al.* [58].

For this, we retrain the ANN model applying only certain subsets of our input features. The prediction power of these models, in this study, portrayed by the mean and median

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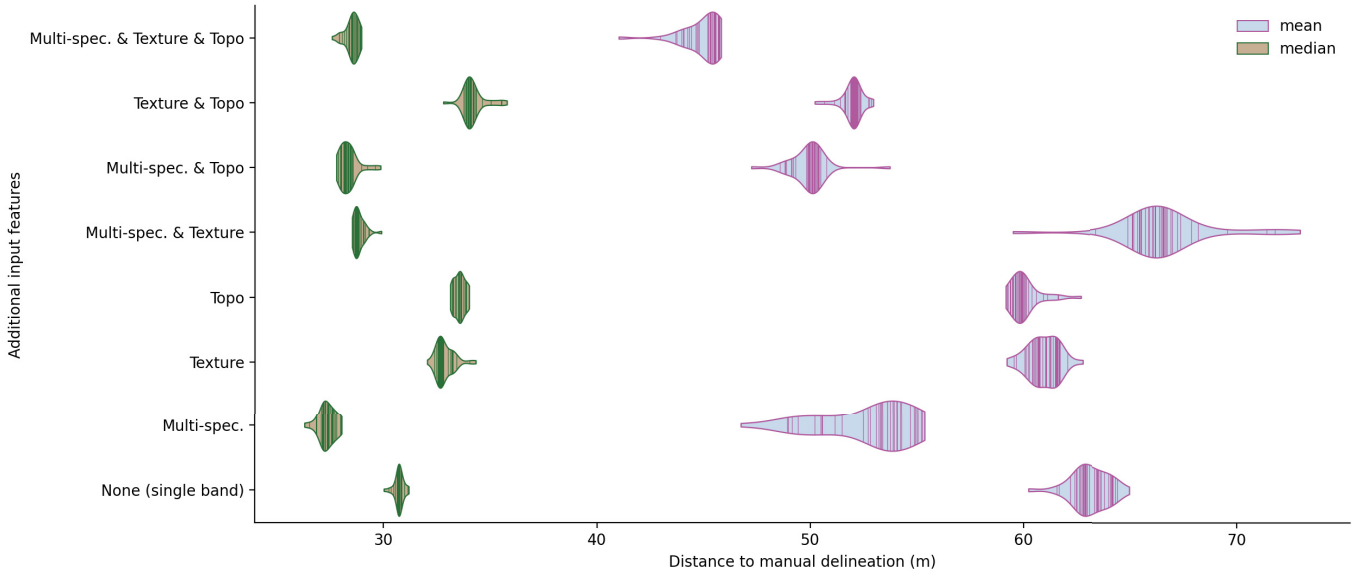


Fig. 7. Results of the feature importance assessment. The various input feature combinations are listed on the Y-axis, with the panchromatic single-band image being supplemented by additional features. The "violin plot" depicts the distribution of mean distance error and median distance error assessed by comparison with the manual calving front delineation. Every vertical line represents one trained model applied on the test dataset. The horizontal extent of the "violin graphs" is defined by corresponding minimum and maximum values and the vertical extent by the frequency distribution.

TABLE I

MEAN AND MEDIAN DISTANCE ERROR AS WELL AS BINARY CLASSIFICATION METRICS ACCURACY AND $F1$-SCORE FOR DIFFERENT COMBINATIONS OF INPUT DATA

Additional input features	Mean distance (m)	Median distance (m)	Binary classification accuracy	F1-Score
Multi-spec. & Texture & Topo	(44.9 ± 0.9)	(28.6 ± 0.5)	(99.10 ± 0.04)	(99.31 ± 0.03)
Texture & Topo	(52.0 ± 0.5)	(34.2 ± 0.5)	(99.08 ± 0.01)	(99.30 ± 0.01)
Multi-spec. & Topo	(49.9 ± 0.9)	(28.3 ± 0.4)	(99.05 ± 0.04)	(99.27 ± 0.03)
Multi-spec. & Texture	(66.5 ± 2.0)	(28.8 ± 0.3)	(98.32 ± 0.07)	(98.74 ± 0.04)
Topo	(60.0 ± 0.7)	(33.5 ± 0.3)	(99.01 ± 0.04)	(99.25 ± 0.02)
Texture	(60.9 ± 0.7)	(32.8 ± 0.4)	(98.47 ± 0.07)	(98.83 ± 0.04)
Multi-spec.	(52.7 ± 2.2)	(27.4 ± 0.4)	(98.68 ± 0.04)	(99.00 ± 0.03)
None (single band)	(63.1 ± 0.9)	(30.7 ± 0.2)	(98.45 ± 0.07)	(98.82 ± 0.06)

distance error to the manual delineation, is then comparatively evaluated. This very pragmatic approach allows us to conclude on the benefit of corresponding omitted or supplemented input features. Rather than evaluating every layer separately, we use the three input datasets introduced in Section II: multispectral bands, topographic model data, and textural features. This is computationally feasible and coherent from a user implementation point of view. Since we want to assess the contribution of each dataset compared with conventional single-band input, the panchromatic band is always included.

ANN training is not deterministic; instead, it is stochastic. Even when using the same training data, every fit model is different and, therefore, performs differently when applied to the test dataset. This is especially prevalent in our workflow, since the loss function used to optimize the model differs from the error metric used in accuracy assessment. A better model convergence could be achieved by applying a custom loss function as well as by implementing additional output layers, which enforce calving front classification more heavily. In this analysis, we enable a thorough comparison between the different input datasets by fitting 40 models for each feature combination. This ensures statistical stability and an estimate for the variability of the predictions.

IV. RESULTS AND DISCUSSION

Fig. 7 presents the outcome of the feature importance analysis. Mean and median distance errors are depicted for various input feature combinations. Each vertical line inside the "violin graphs" represents one trained model applied to the test dataset. Table I lists the corresponding statistics. It also quotes the binary classification metrics accuracy and $F1$-score for comparability to the existing studies [31], [32]. Regardless of the applied feature set, the median error is always smaller than the mean error. This is due to some challenging test scenes, which exhibit large deviations to the manual delineation. The mean error characterizes the predictive power of a model under these conditions. When using the panchromatic band only, without additional input features, the trained models provide reliable predictions with 63-m mean and 31-m median error for our test dataset (last line in Fig. 7 and Table I). Additional input features can both increase and decrease a model's predictive power. For the median error, an improvement in model performance with respect to

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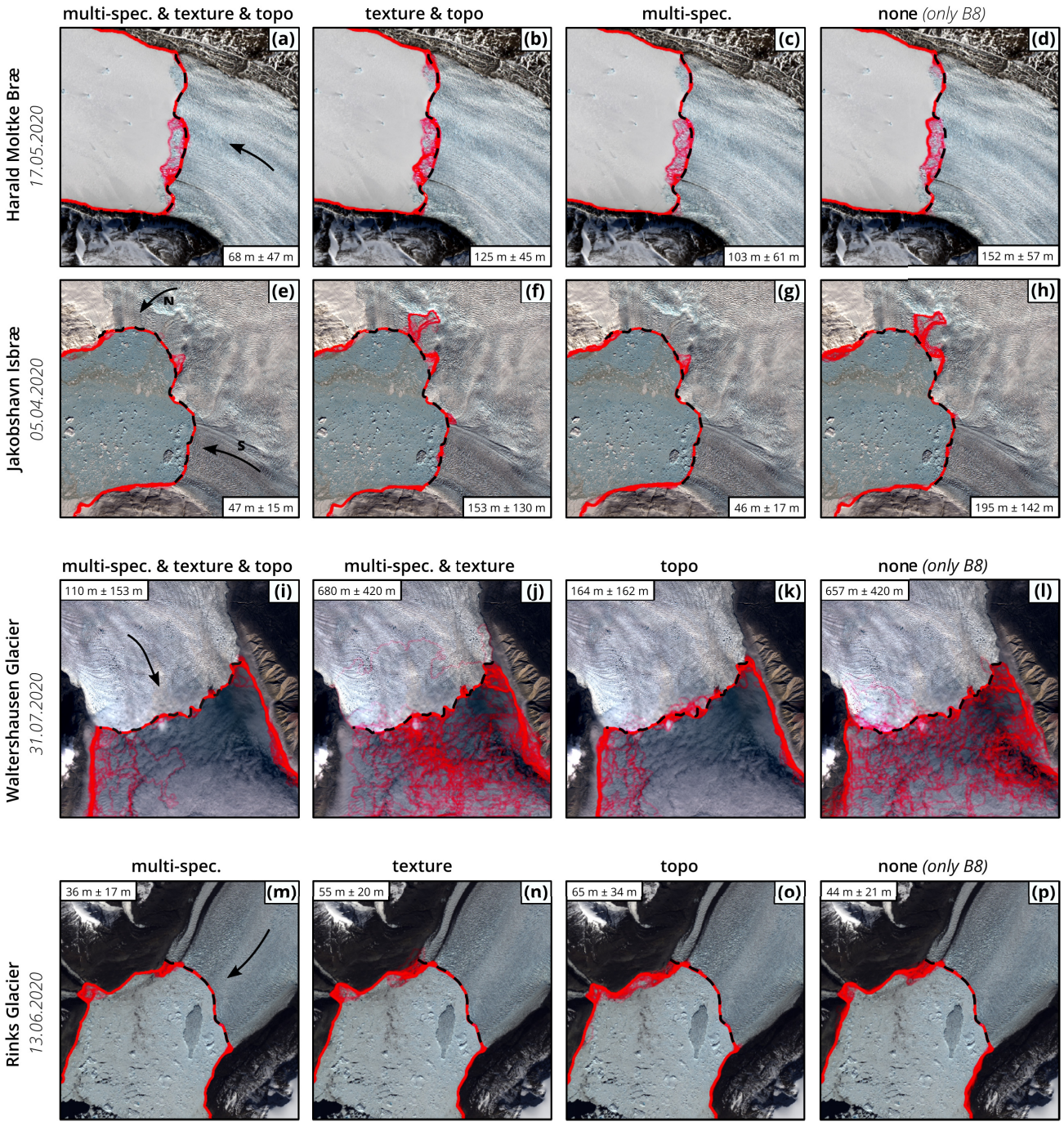


Fig. 8. Feature importance exemplarily shown for four different test scenes. Shown are RGB images with ground truth as a black dashed line, ice flow direction as an arrow, and the 40 model predictions in semitransparent red for different glaciers (rows) and different input features denoted above the respective columns. The mean distance error for the depicted predictions at the glacier front is also given.

conventional single-band processing is only observed when applying multispectral features. In contrast, the median error increases if textural and/or topography data are added without adding multispectral features. This indicates that both of these datasets can increase the models susceptibility for overfitting, which lead to systematic misclassifications at the calving front throughout the whole test dataset. Notably, models using only multispectral features have a slightly better median performance compared with models using all available data. Evaluating the mean distance error of our models enables an assessment on the handling of challenging scenes, as the mean is particularly sensitive toward outliers. With the exception of combining multispectral with textural inputs, all assessed models exhibit an improvement in mean error with respect to the single-band processing. Combining all three presented datasets results in a significant mean error decrease from 63 to 45 m, emphasizing their benefit for processing imagery under challenging conditions.

Fig. 8 presents the specific examples of the above-concluded characteristics. Each of the 40 model predictions is shown in

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semi-transparent red with increasing intensity the more the predictions overlap. The manually delineated front position is drawn as a black dashed line. The presence of fast ice usually leads to abrupt textural changes ahead of the actual glacier front. This is a result of the fast ice breaking at the glacier front or calved ice, which is not transported away. Conditions like these often lead to systematic misclassifications of the ANN, since both gray value and texture help very little in differentiating these surfaces. This causes the predicted front to follow the wrong textural transition and results in a large distance error. The case of Harald Moltke Bræ [Fig. 8(a)–(d)] demonstrates these characteristics and confirms a significant accuracy increase by including multispectral information. This is a result of varying reflectivities of different ice, snow, and ice-mélange surfaces. Fig. 8(e)–(h), showing a scene of Jakobshavn Isbræ, manifests the same aspect in a slightly different way. Frequent breakups of the northern branch terminus lead to an unpronounced textural transition at the calving front. Without utilizing multispectral information, these scenes remain very hard to delineate. The third scene [Fig. 8(i)–(l)] shows the performance using different input feature sets at Waltershausen Glacier. Thick clouds cover most of the scene and barely allow us to identify the calving front. Combined with the fact that this glacier is not part of the training dataset, this scene is very challenging to process by the ANN. The different models clearly show that under conditions like these, stable predictions are only possible using topography data. In particular, the calving front—coastline transition as well as large parts of the coastline itself are not visible in this scene, thus making external data mandatory for an accurate delineation. Topographic information provides an important constraint of the calving front position inside the fjord. Finally, the case of Rinks Glacier [Fig. 8(m)–(p)] is representative of a large portion of the test imagery, which is reliably delineated, with distance errors well below 100 m, regardless of the used input information. Still, we notice a decreased distance error when applying multispectral data. Interestingly, but also in accordance with our findings in Fig. 7, the addition of textural as well as topography data increases the distance error. This happens for a large number of otherwise easy to delineate test scenes, and mostly for certain parts of the glacier front with changing ice-mélange texture or pronounced glacial meltwater plumes. Hence, we suspect increased overfitting when including textural and topography data, which manifests in a larger median error over the whole dataset. Although we applied several techniques to counteract model overfitting, modifying the ANN architecture, specifically adjusting its complexity, might be necessary to take full advantage of these input features.

V. Summary and Outlook

Our results provide important insights for advancing ANN-based calving front extraction. The feature importance portrayed in this study emphasizes both the potential in integrating additional input information as well as the significance of their thoughtful selection. The integration of multispectral bands is not only convenient, since most data products deliver this information anyway, but also leads to more accurate predictions especially under challenging conditions. Using topography and textural inputs cannot be recommended without reservation. Both datasets can help for certain scenes, but result in a higher median distance error for our test dataset, which indicates that the corresponding models are overfitted. The integration of these datasets needs to be done carefully and, if possible, accompanied by further investigations and model adjustments.

The results of this assessment are not only relevant for advancing automated calving front extraction, but also for a wider range of glaciological land surface classification tasks using optical imagery and deep neural networks. Recent studies already explored the application of deep learning algorithms for mapping supraglacial lakes and debris-covered glaciers. The results presented in our contribution reinforce these existing efforts but also lay the foundation for further applications and developments.

Code and Data Availability

The code used for preprocessing, ANN training, and accuracy assessment as well as all reference data applied in this study are available at the TU Dresden Open Access Repository and Archive (http://dx.doi.org/10.25532/OPARA-183). In particular, this includes 728 manually delineated calving front positions, which we provide in a georeferenced shapefile format, as well as 995 machine learning ready input raster subsets with their corresponding, manual delineated, segmentation mask.

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